

Problem 3: Way Prediction on an L1 Instruction Cache (8 points)

Throughout, we use the standard average-memory-access-time expression, in which the miss penalty is an *additional* cost incurred on top of the access time:

$$\text{AMAT} = \text{access time} + \text{miss rate} \times \text{miss penalty}. \quad (1)$$

The stated geometry (4-way set associative, 64 B blocks, 256 KB capacity) fixes the baseline 1.1 ns access time and the direct-mapped 1 ns comparison point; it is not needed numerically below.

Question A (2 points)

With way prediction the cache first probes the single predicted way. If that prediction is correct the access completes in the direct-mapped access time of 1 ns. If it is wrong, the 1.1 ns quoted for a misprediction is *additional* time, so the failed probe is followed by the ordinary associative lookup and the two costs add:

$$t_{\text{best}} = 1 \text{ ns}, \quad (2)$$

$$t_{\text{worst}} = 1 + 1.1 = 2.1 \text{ ns}. \quad (3)$$

Answer. For the best case a cache access takes **1 ns**; for the worst case it takes **2.1 ns**.

Here “best/worst case” is read as the two *cache-access* (hit-time) outcomes of the way predictor, namely prediction success versus prediction failure, since the 50 ns miss penalty is accounted separately in Eq. (1) rather than as part of the access time.

Question B (6 points)

Step 1: baseline AMAT (no way prediction). The original 4-way cache has a 1.1 ns access time, a 1% miss rate and a 50 ns additional miss penalty:

$$\text{AMAT}_{\text{base}} = 1.1 + 0.01 \times 50 = 1.1 + 0.5 = 1.6 \text{ ns}. \quad (4)$$

Step 2: AMAT with way prediction. Let a be the prediction accuracy. The prediction succeeds with probability a (costing $t_{\text{best}} = 1$ ns) and fails with probability $1 - a$ (costing $t_{\text{worst}} = 2.1$ ns), so the expected access time is $a(1) + (1 - a)(2.1)$. Since the miss rate is unchanged at 1% and the miss penalty is still 50 ns, the miss term is identical to the baseline:

$$\begin{aligned} \text{AMAT}_{\text{wp}}(a) &= [a \times 1 + (1 - a) \times 2.1] + 0.01 \times 50 \\ &= (2.1 - 1.1a) + 0.5 \\ &= 2.6 - 1.1a \text{ ns}. \end{aligned} \quad (5)$$

Step 3: impose the 5% reduction. Reducing the AMAT by at least 5% means the new AMAT is at most 95% of the baseline:

$$\text{AMAT}_{\text{wp}}(a) \leq 0.95 \times \text{AMAT}_{\text{base}} = 0.95 \times 1.6 = 1.52 \text{ ns}. \quad (6)$$

Step 4: solve for a . Substituting Eq. (5) into Eq. (6):

$$\begin{aligned} 2.6 - 1.1a &\leq 1.52 \\ 1.1a &\geq 2.6 - 1.52 = 1.08 \\ a &\geq \frac{1.08}{1.1} = \frac{108}{110} = \frac{54}{55} = 0.9818\overline{18}. \end{aligned} \quad (7)$$

Answer. The prediction accuracy must be *at least*

$$a_{\min} = \frac{54}{55} = 98.\overline{18}\% \approx \mathbf{98.19\%}. \quad (8)$$

Because this is a lower bound, the decimal answer is rounded *up*: the exact threshold $98.1818\dots\%$ rounded down to 98.18% would give $\text{AMAT}_{\text{wp}} = 2.6 - 1.1(0.9818) = 1.52002\text{ ns}$, which exceeds the 1.52 ns budget and therefore fails the requirement, whereas 98.19% gives 1.51991 ns and satisfies it.

Check. At $a = 54/55$ exactly, $\text{AMAT}_{\text{wp}} = 2.6 - 1.1 \times \frac{54}{55} = 2.6 - 1.08 = 1.52\text{ ns}$, which is exactly $0.95 \times 1.6\text{ ns}$, i.e. precisely a 5% reduction, as required at the threshold. Note also that even a perfect predictor ($a = 1$) only reaches $\text{AMAT} = 1.5\text{ ns}$, a 6.25% reduction, so the 5% target leaves very little margin — which is why the required accuracy is so high.